Exercises: Statistical Testing and Confidence Regions – Answer Key

Mathematical Foundations for Machine Learning

Statistical Testing Fundamentals

Exercise 1.1 [Easy]

A medical test is used to detect a disease. Let H_0: "patient is healthy" and H_1: "patient is sick." The test is applied to 200 patients and produces the following results:

	Test says Healthy	Test says Sick
Truly Healthy	130	20
Truly Sick	10	40

(a) Identify the Type I error and the Type II error in this context. Which one corresponds to a false positive?

(b) Compute the significance level (Type I error rate) and the Type II error rate from the table.

(c) Compute the power of this test. Interpret the result in one sentence.

Solution

(a) **Type I error** (false positive): rejecting H_0 when it is true, i.e., declaring a healthy patient sick. **Type II error** (false negative): failing to reject H_0 when H_1 is true, i.e., declaring a sick patient healthy. The Type I error corresponds to a false positive.

(b) There are $130 + 20 = 150$ truly healthy patients. Of these, 20 are incorrectly classified as sick.

$$\text{Type I error rate (significance level)} = \alpha = \frac{20}{150} = \frac{2}{15} \approx 0.133.$$

There are $10 + 40 = 50$ truly sick patients. Of these, 10 are incorrectly classified as healthy.

$$\text{Type II error rate} = \beta = \frac{10}{50} = 0.2.$$

(c) The power of the test is

$$\text{Power} = 1 - \beta = 1 - 0.2 = 0.8.$$

This means that if a patient truly has the disease, the test correctly detects it 80% of the time.

Exercise 1.2 [Easy]

Let $X_1, \ldots, X_n$ be i.i.d. samples from a distribution with mean μ and known variance σ^2. We construct the confidence interval

$$C_n = \left[\bar{X}_n - \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2},\ \bar{X}_n + \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2}\right],$$

where $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ and q_γ is the γ-quantile of $\mathcal{N}(0,1)$.

(a) State precisely: what is random in C_n, and what is fixed? Is the statement "μ falls in C_n with probability $1-\alpha$" correct?

(b) Suppose $\sigma = 2$, $\alpha = 0.05$, and $q_{0.975} = 1.96$. How large must n be so that the width of the confidence interval is at most 0.5?

(c) A colleague claims: "If I compute a 95% confidence interval and the parameter μ is not in it, then my experiment failed." Is this interpretation correct? Explain briefly.

Solution

(a) The interval C_n is random (it depends on the data $X_1, \ldots, X_n$ through $\bar{X}_n$), while the true parameter μ is a fixed unknown constant. The statement is correct in the frequentist sense: if we repeat the experiment many times and construct C_n each time, the true μ will be contained in approximately $(1-\alpha) \cdot 100\%$ of the intervals.

(b) The width of the confidence interval is $2 \cdot \frac{\sigma}{\sqrt{n}} q_{0.975}$. We need

$$2 \cdot \frac{2}{\sqrt{n}} \cdot 1.96 \leq 0.5 \implies \frac{7.84}{\sqrt{n}} \leq 0.5 \implies \sqrt{n} \geq 15.68 \implies n \geq 245.9.$$

So we need $n \geq 246$.

(c) The interpretation is incorrect. A 95% confidence interval means that 5% of all such intervals (across repeated experiments) will not contain μ. Not containing μ in a particular experiment is expected to happen with probability $\alpha = 0.05$ and does not indicate a failed experiment.

Exercise 1.3 [Medium]

A classifier is evaluated on a test set of n samples. Each sample is independently classified correctly with unknown probability p (the true accuracy). Let $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} Z_i$ be the empirical accuracy, where $Z_i \in \{0, 1\}$ indicates whether sample i was correctly classified.

(a) Compute $\mathbb{E}[\hat{p}]$ and $\mathrm{Var}(\hat{p})$.

(b) Using Chebyshev's inequality, show that for any $\varepsilon > 0$,

$$P(|\hat{p} - p| \geq \varepsilon) \leq \frac{1}{4n\varepsilon^2}.$$

Hint: What is the maximum of $p(1-p)$ over $p \in [0, 1]$?

(c) Using the result from (b), how many test samples n are needed to guarantee that $|\hat{p} - p| < 0.05$ with probability at least 95%?

Solution

(a) Since $Z_i \sim \mathrm{Bernoulli}(p)$ are i.i.d.:

$$\mathbb{E}[\hat{p}] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Z_i] = p.$$

$$\mathrm{Var}(\hat{p}) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(Z_i) = \frac{p(1-p)}{n}.$$

(b) We apply Chebyshev's inequality to $\hat{p}$. Chebyshev states $P(|Y - \mathbb{E}[Y]| \geq k) \leq \frac{\mathrm{Var}(Y)}{k^2}$. Mapping our variables: $Y = \hat{p}$, $\mathbb{E}[Y] = p$, $k = \varepsilon$, $\mathrm{Var}(Y) = \frac{p(1-p)}{n}$. Thus:

$$P(|\hat{p} - p| \geq \varepsilon) \leq \frac{p(1-p)}{n\varepsilon^2}.$$

Since $p(1-p) \leq \frac{1}{4}$ for all $p \in [0, 1]$ (the maximum of $g(p) = p(1-p)$ is at $p = 1/2$ where

$g(1/2) = 1/4$), we obtain

$$P(|\hat{p} - p| \geq \varepsilon) \leq \frac{1}{4n\varepsilon^2}.$$

(c) We need $P(|\hat{p} - p| \geq 0.05) \leq 0.05$. From (b):

$$\frac{1}{4n(0.05)^2} \leq 0.05 \implies \frac{1}{0.01n} \leq 0.05 \implies n \geq \frac{1}{0.01 \cdot 0.05} = 2000.$$

We need at least $n = 2000$ test samples.

Exercise 1.4 [Medium]

Consider a hypothesis test with null hypothesis $H_0 : \theta = \theta_0$ and a test statistic $T = T(X_1, \ldots, X_n)$ whose distribution under H_0 is known. The p-value is defined as

$$\text{p-value} = P_{H_0}(T \geq t_{\text{obs}}),$$

where t_{obs} is the observed value of T.

(a) A researcher runs a test and obtains a p-value of 0.03. The significance level is $\alpha = 0.05$. State the decision rule and whether H_0 is rejected.

(b) The researcher repeats the same experiment independently 5 times and obtains p-values: $0.03, 0.12, 0.07, 0.04, 0.15$. If the significance level is $\alpha = 0.05$ for each individual test, a colleague argues that at least one rejection is "proof" that H_0 is false. Show that even if H_0 is true, the probability of at least one rejection among m independent tests at level α equals $1 - (1 - \alpha)^m$, and compute this for $m = 5$, $\alpha = 0.05$.

(c) Explain how the Bonferroni correction addresses the issue in (b). What adjusted significance level should be used for each of the 5 tests to control the overall false rejection probability at $\alpha = 0.05$?

Solution

(a) Decision rule: reject H_0 if p-value $\leq \alpha$. Since $0.03 \leq 0.05$, we reject H_0.

(b) Under H_0, each test independently rejects with probability α. Let R_i be the event that test i rejects. Then

$$P(\text{at least one rejection}) = 1 - P(\text{no rejections}) = 1 - \prod_{i=1}^{m} P(R_i^c) = 1 - (1 - \alpha)^m.$$

For $m = 5$, $\alpha = 0.05$:

$$1 - (1 - 0.05)^5 = 1 - (0.95)^5 = 1 - 0.7738 \approx 0.226.$$

So there is roughly a 22.6% chance of at least one false rejection, which is far larger than the nominal 5%.

(c) The Bonferroni correction replaces the individual significance level α with $\frac{\alpha}{m}$. That is, we reject H_0^i only if $p_i \leq \frac{\alpha}{m}$. By the union bound,

$$P(\text{at least one false rejection}) = P\left(\bigcup_{i=1}^{m} R_i\right) \leq \sum_{i=1}^{m} P(R_i) = m \cdot \frac{\alpha}{m} = \alpha.$$

For $m = 5$ and $\alpha = 0.05$, the adjusted level is $\frac{0.05}{5} = 0.01$. Under this correction, among the 5 p-values $(0.03, 0.12, 0.07, 0.04, 0.15)$, none is ≤ 0.01, so no test rejects.

Exercise 1.5 [Hard]

Let T be a continuous test statistic and define the p-value as $p = P_{H_0}(T \geq T_{\text{obs}})$, where T_{obs} is the observed realization of T.

(a) Let F_0 denote the CDF of T under H_0. Show that the p-value can be written as $p = 1 - F_0(T_{\text{obs}})$.

(b) Prove that under H_0, the p-value is uniformly distributed on $[0, 1]$, i.e., $p \sim \text{Uniform}(0, 1)$.

Hint: Use the probability integral transform: if T has CDF F_0 and F_0 is continuous, then $F_0(T) \sim \text{Uniform}(0, 1)$.

(c) Using the result of (b), show that a test that rejects H_0 when $p \leq \alpha$ has exactly significance level α, i.e., $P_{H_0}(p \leq \alpha) = \alpha$.

Solution

(a) Since $F_0(t) = P_{H_0}(T \leq t)$, we have

$$p = P_{H_0}(T \geq T_{\text{obs}}) = 1 - P_{H_0}(T < T_{\text{obs}}) = 1 - F_0(T_{\text{obs}}),$$

where the last step uses continuity of T so that $P(T = T_{\text{obs}}) = 0$.

(b) We want to show that $p = 1 - F_0(T)$ is uniform on $[0, 1]$. We compute the CDF of p: for $u \in [0, 1]$,

$$P_{H_0}(p \leq u) = P_{H_0}(1 - F_0(T) \leq u) = P_{H_0}(F_0(T) \geq 1 - u).$$

By the probability integral transform, $U := F_0(T) \sim \text{Uniform}(0, 1)$ under H_0. Therefore

$$P_{H_0}(F_0(T) \geq 1 - u) = P(U \geq 1 - u) = 1 - (1 - u) = u.$$

Since $P_{H_0}(p \leq u) = u$ for all $u \in [0, 1]$, we conclude $p \sim \text{Uniform}(0, 1)$.

(c) If $p \sim \text{Uniform}(0, 1)$ under H_0, then

$$P_{H_0}(p \leq \alpha) = \alpha$$

directly from the CDF of the uniform distribution. Hence rejecting when $p \leq \alpha$ gives a test with exact significance level α.

Confidence Intervals and the Gauss Test

Exercise 2.1 [Easy]

Consider i.i.d. samples $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ with known σ^2. We wish to test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ using the Gauss test statistic

$$Z = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma}.$$

(a) State the distribution of Z under H_0.

(b) For a two-sided test at significance level α, state the rejection region in terms of Z and the quantile $q_{1-\alpha/2}$ of $\mathcal{N}(0, 1)$.

(c) Suppose $n = 25$, $\sigma = 3$, $\bar{X}_n = 5.2$, $\mu_0 = 4.5$, and $q_{0.975} = 1.96$. Compute Z and state whether H_0 is rejected at $\alpha = 0.05$.

Solution

(a) Under $H_0 : \mu = \mu_0$, we have $\bar{X}_n \sim \mathcal{N}(\mu_0, \sigma^2/n)$, so

$$Z = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} \sim \mathcal{N}(0, 1).$$

(b) Reject H_0 if $|Z| > q_{1-\alpha/2}$, i.e., the rejection region is $\{Z : |Z| > q_{1-\alpha/2}\}$.
(c)

$$Z = \frac{\sqrt{25}(5.2 - 4.5)}{3} = \frac{5 \cdot 0.7}{3} = \frac{3.5}{3} \approx 1.167.$$

Since $|Z| = 1.167 < 1.96 = q_{0.975}$, we do not reject H_0 at $\alpha = 0.05$.

Exercise 2.2 [Medium]

Consider the Gauss test setting: $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d. with known σ^2, and the two-sided test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$ at level α.

(a) Show that the test "reject H_0 if $|Z| > q_{1-\alpha/2}$" is equivalent to "reject H_0 if $\mu_0 \notin C_n$", where

$$C_n = \left[\bar{X}_n - \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2},\ \bar{X}_n + \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2}\right]$$

is the $(1 - \alpha)$-confidence interval for μ.

(b) This shows that hypothesis tests and confidence intervals are two sides of the same coin. Formulate the general principle: a value θ_0 is rejected by a level-α test if and only if $\theta_0 \notin \ldots$ (complete the statement).

(c) Suppose we are comparing two classifiers A and B on a test set of n samples. Classifier A has accuracy $\hat{p}_A = 0.82$ and classifier B has $\hat{p}_B = 0.79$, each evaluated on independent test sets of size $n = 400$. Using the normal approximation, construct a 95% confidence interval for the difference $p_A - p_B$. Does the interval contain 0? Interpret the result.

Solution

(a) $|Z| > q_{1-\alpha/2}$ means

$$\left|\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma}\right| > q_{1-\alpha/2} \iff |\bar{X}_n - \mu_0| > \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2}.$$

This is equivalent to $\mu_0 \notin [\bar{X}_n - \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2},\ \bar{X}_n + \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2}] = C_n$. Hence the two formulations are equivalent.
(b) A value θ_0 is rejected by a level-α test if and only if $\theta_0 \notin C_n$, where C_n is the corresponding $(1 - \alpha)$-confidence interval (or confidence set) for the parameter.
(c) For independent test sets, $\hat{p}_A - \hat{p}_B$ has approximate variance $\frac{\hat{p}_A(1-\hat{p}_A)}{n} + \frac{\hat{p}_B(1-\hat{p}_B)}{n}$. Plugging in:

$$\hat{\sigma}^2 = \frac{0.82 \cdot 0.18}{400} + \frac{0.79 \cdot 0.21}{400} = \frac{0.1476}{400} + \frac{0.1659}{400} = \frac{0.3135}{400} = 0.00078375.$$

So $\hat{\sigma} \approx 0.028$. The difference is $\hat{p}_A - \hat{p}_B = 0.03$. The 95% confidence interval is

$$0.03 \pm 1.96 \cdot 0.028 = 0.03 \pm 0.055 = [-0.025, 0.085].$$

The interval contains 0, so we cannot reject at the 5% level the hypothesis that classifiers A and B have equal accuracy. Despite the observed 3 percentage point difference, it is not statistically significant at this sample size.

Exercise 2.3 [Hard]

The power function of a test is defined as $\beta(\theta) = P_\theta(\text{reject } H_0)$.

Consider the one-sided Gauss test: $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d. with known σ^2, testing $H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$. The test rejects H_0 if $Z = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} > q_{1-\alpha}$.

(a) Derive the power function $\beta(\mu)$ for this test. Express it in terms of the standard normal CDF Φ.

Hint: Under μ, write $Z = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} + \frac{\sqrt{n}(\mu - \mu_0)}{\sigma}$.

(b) Show that $\beta(\mu_0) = \alpha$ and that $\beta(\mu) \to 1$ as $\mu \to \infty$. What does this mean about the test?

(c) Suppose $\sigma = 1$, $\alpha = 0.05$ (so $q_{0.95} \approx 1.645$), and we want the test to have power at least 0.9 when $\mu = \mu_0 + 0.5$. Find the minimum sample size n.

Hint: $\Phi^{-1}(0.9) \approx 1.282$.

Solution

(a) Under the true parameter μ, $\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$, so

$$Z = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} = \underbrace{\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}}_{\sim \mathcal{N}(0,1)} + \frac{\sqrt{n}(\mu - \mu_0)}{\sigma}.$$

Let $\delta = \frac{\sqrt{n}(\mu - \mu_0)}{\sigma}$. Then $Z \sim \mathcal{N}(\delta, 1)$. The power is:

$$\beta(\mu) = P_\mu(Z > q_{1-\alpha}) = P(\mathcal{N}(\delta, 1) > q_{1-\alpha}) = 1 - \Phi(q_{1-\alpha} - \delta).$$

Substituting δ:

$$\beta(\mu) = 1 - \Phi\left(q_{1-\alpha} - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma}\right).$$

(b) At $\mu = \mu_0$: $\delta = 0$, so $\beta(\mu_0) = 1 - \Phi(q_{1-\alpha}) = 1 - (1 - \alpha) = \alpha$. This confirms the test has the correct significance level. As $\mu \to \infty$: $\delta \to \infty$, so $q_{1-\alpha} - \delta \to -\infty$ and $\Phi(q_{1-\alpha} - \delta) \to 0$, giving $\beta(\mu) \to 1$. This means the test is *consistent*: it eventually detects any deviation from H_0 with probability approaching 1.

(c) We need $\beta(\mu_0 + 0.5) \geq 0.9$:

$$1 - \Phi\left(q_{0.95} - \frac{\sqrt{n} \cdot 0.5}{1}\right) \geq 0.9 \implies \Phi\big(1.645 - 0.5\sqrt{n}\big) \leq 0.1.$$

Since $\Phi^{-1}(0.1) = -\Phi^{-1}(0.9) \approx -1.282$:

$$1.645 - 0.5\sqrt{n} \leq -1.282 \implies 0.5\sqrt{n} \geq 2.927 \implies \sqrt{n} \geq 5.854 \implies n \geq 34.3.$$

We need $n \geq 35$.

Permutation Tests

Exercise 3.1 [Easy]

Suppose we have two groups of observations: $X_1 = 3, X_2 = 7$ and $Y_1 = 10$. We test $H_0 : F_X = F_Y$ (the two groups come from the same distribution) using the test statistic $T = |\bar{X} - \bar{Y}|$.

(a) Compute the observed test statistic t_{obs}.

(b) List all $\binom{3}{2} = 3$ ways to split the pooled data $\{3, 7, 10\}$ into a group of size 2 and a group of size 1, and compute T for each split.

(c) Compute the p-value, i.e., $P_0(T \geq t_{\text{obs}})$, by counting the fraction of permutations where $T \geq t_{\text{obs}}$. Can we reject H_0 at $\alpha = 0.05$?

Solution

(a) $\bar{X} = \frac{3+7}{2} = 5$, $\bar{Y} = 10$, so $t_{\text{obs}} = |5 - 10| = 5$.
(b) All splits of $\{3, 7, 10\}$ into (group of 2, group of 1):

Split	Group means	$T = \lvert\bar{X} - \bar{Y}\rvert$
$\{3, 7\}$ vs $\{10\}$	5 vs 10	5
$\{3, 10\}$ vs $\{7\}$	6.5 vs 7	0.5
$\{7, 10\}$ vs $\{3\}$	8.5 vs 3	5.5

(c) The p-value is the fraction of splits where $T \geq t_{\text{obs}} = 5$. We have $T = 5, 0.5, 5.5$. Two out of three splits satisfy $T \geq 5$:

$$\text{p-value} = \frac{2}{3} \approx 0.667.$$

Since $0.667 > 0.05$, we cannot reject H_0. With such a tiny sample, there is not enough evidence.

Exercise 3.2 [Medium]

In practice, the exact permutation test is intractable for large datasets because the number of permutations $\binom{N}{m}$ grows combinatorially. Instead, one uses a **Monte Carlo approximation**: sample S random permutations and estimate

$$\hat{p} = \frac{1}{S} \sum_{j=1}^{S} \mathbf{1}_{T_j \geq t_{\text{obs}}}.$$

(a) Show that $\hat{p}$ is an unbiased estimator of the true p-value $p^* = P_0(T \geq t_{\text{obs}})$.

(b) Compute $\text{Var}(\hat{p})$ and show it is at most $\frac{1}{4S}$.

(c) Using Chebyshev's inequality, how many Monte Carlo samples S are needed to guarantee $|\hat{p} - p^*| < 0.01$ with probability at least 0.99?

Solution

(a) Each $\mathbf{1}_{T_j \geq t_{\text{obs}}}$ is a Bernoulli random variable with parameter p^* (since each permutation is drawn uniformly from the permutation distribution). Therefore

$$\mathbb{E}[\hat{p}] = \frac{1}{S} \sum_{j=1}^{S} \mathbb{E}[\mathbf{1}_{T_j \geq t_{\text{obs}}}] = \frac{1}{S} \cdot S \cdot p^* = p^*.$$

So $\hat{p}$ is unbiased.
(b) Since the S random permutations are drawn independently,

$$\text{Var}(\hat{p}) = \frac{1}{S^2} \sum_{j=1}^{S} \text{Var}(\mathbf{1}_{T_j \geq t_{\text{obs}}}) = \frac{p^*(1 - p^*)}{S}.$$

Since $p^*(1 - p^*) \leq \frac{1}{4}$, we have $\text{Var}(\hat{p}) \leq \frac{1}{4S}$.

(c) Applying Chebyshev to $\hat{p}$ with $\mathbb{E}[\hat{p}] = p^*$ and $\mathrm{Var}(\hat{p}) \leq \frac{1}{4S}$:

$$P(|\hat{p} - p^*| \geq 0.01) \leq \frac{1/(4S)}{(0.01)^2} = \frac{1}{4S \cdot 10^{-4}} = \frac{2500}{S}.$$

We need this to be ≤ 0.01:

$$\frac{2500}{S} \leq 0.01 \implies S \geq 250000.$$

At least $S = 250{,}000$ Monte Carlo samples are needed.

Mixed Challenging Exercises

Exercise 4.1 [Hard]

A machine learning practitioner trains a binary classifier and evaluates it on a test set of $n = 500$ i.i.d. samples. The classifier achieves an empirical accuracy of $\hat{p} = 0.74$.

(a) The practitioner wants to test $H_0 : p \leq 0.70$ (the classifier is no better than 70% accuracy) against $H_1 : p > 0.70$. Using the normal approximation under H_0 with worst-case variance (i.e., $p_0 = 0.70$), compute the test statistic Z and decide at $\alpha = 0.05$ ($q_{0.95} \approx 1.645$).

(b) Construct a 95% confidence interval for p using the normal approximation with estimated variance $\hat{p}(1 - \hat{p})/n$. Verify that the decision from part (a) is consistent with whether 0.70 lies inside or outside this interval.

(c) The practitioner now evaluates the same classifier on $k = 10$ different test sets (e.g., different demographic subgroups) and tests $H_0^i : p_i \leq 0.70$ for each. Explain why a Bonferroni correction is appropriate and compute the adjusted significance level. If the smallest p-value across the 10 tests is 0.008, is any subgroup's performance significantly above 70%?

Solution

(a) Under H_0 with $p_0 = 0.70$:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0.74 - 0.70}{\sqrt{0.70 \cdot 0.30/500}} = \frac{0.04}{\sqrt{0.00042}} = \frac{0.04}{0.02049} \approx 1.952.$$

Since $Z = 1.952 > 1.645 = q_{0.95}$, we reject H_0 at $\alpha = 0.05$. There is significant evidence that the true accuracy exceeds 70%.

(b) Using estimated variance:

$$\hat{\sigma} = \sqrt{\frac{0.74 \cdot 0.26}{500}} = \sqrt{0.000384} \approx 0.01962.$$

The 95% confidence interval is

$$\hat{p} \pm 1.96 \cdot \hat{\sigma} = 0.74 \pm 1.96 \cdot 0.01962 = 0.74 \pm 0.0385 = [0.7015,\ 0.7785].$$

Since $0.70 < 0.7015$, the value 0.70 lies below (just outside) the confidence interval. This is consistent with the rejection decision in (a): values outside the confidence region are rejected by the corresponding test.

(c) When testing $k = 10$ hypotheses simultaneously, performing each test at level $\alpha = 0.05$ inflates the probability of at least one false rejection well beyond 5%. The Bonferroni correction controls the family-wise error rate by using an adjusted significance level of $\frac{\alpha}{k} = \frac{0.05}{10} = 0.005$ for each individual test. The smallest p-value is $0.008 > 0.005$, so after Bonferroni correction, no subgroup's accuracy is found to be significantly above 70%.

Exercise 4.2 [Hard]

Let $X_1, \ldots, X_n$ be i.i.d. with $X_i \in [0,1]$ (e.g., bounded loss values of a model). Let $\mu = \mathbb{E}[X_i]$ denote the true expected loss.

(a) Using Hoeffding's inequality, show that

$$P\big(|\bar{X}_n - \mu| \geq t\big) \leq 2\exp(-2nt^2).$$

Recall Hoeffding: For independent $X_i \in [a_i, b_i]$, $P\big(|\bar{X}_n - \mu| \geq t\big) \leq 2\exp\Big(\frac{-2n^2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\Big)$.

(b) Invert the inequality from (a) to construct a $(1-\alpha)$-confidence interval for μ that does not require knowledge of the variance. Express the interval width as a function of n and α.

(c) Compare the width of this Hoeffding-based confidence interval to the normal-approximation confidence interval $\bar{X}_n \pm \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2}$ (assuming σ known). For which regime (small n or large n) is each preferable, and why?

Solution

(a) We apply Hoeffding's inequality with $a_i = 0$, $b_i = 1$ for all i. Then $\sum_{i=1}^n (b_i - a_i)^2 = n \cdot 1^2 = n$. Substituting:

$$P(|\bar{X}_n - \mu| \geq t) \leq 2\exp\left(\frac{-2n^2t^2}{n}\right) = 2\exp(-2nt^2).$$

(b) We want the right-hand side to equal α: set $2\exp(-2nt^2) = \alpha$, which gives

$$t = \sqrt{\frac{\log(2/\alpha)}{2n}}.$$

The confidence interval is therefore

$$C_n = \left[\bar{X}_n - \sqrt{\frac{\log(2/\alpha)}{2n}},\ \bar{X}_n + \sqrt{\frac{\log(2/\alpha)}{2n}}\right].$$

The total width is $2\sqrt{\frac{\log(2/\alpha)}{2n}}$, which depends only on n and α (no variance needed).

(c) The Hoeffding-based interval has half-width $\propto \frac{1}{\sqrt{n}}\sqrt{\log(2/\alpha)}$, while the normal interval has half-width $\propto \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2}$. Key differences:

Small n: The Hoeffding interval is preferable because it is a *finite-sample* bound — it holds exactly for any n without requiring the Central Limit Theorem approximation. The normal interval relies on the CLT, which may be inaccurate for small n.

Large n: The normal interval is preferable because it adapts to the true variance σ^2, which is typically much smaller than the worst-case bound of $1/4$ implicit in Hoeffding. For example, if X_i has small variance, the normal interval shrinks accordingly, while the Hoeffding interval only uses the range $[0,1]$ and is therefore unnecessarily wide.